

## Odd End Semester Examination December 2025

Name of the Program: B. Tech  
Name of the Course: Basic Electrical Engineering

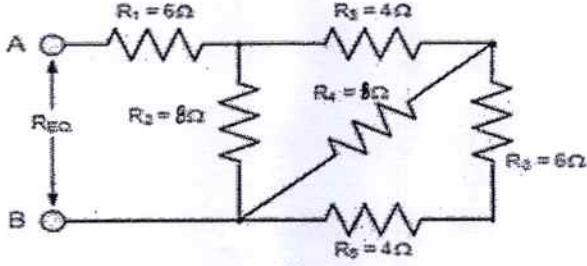
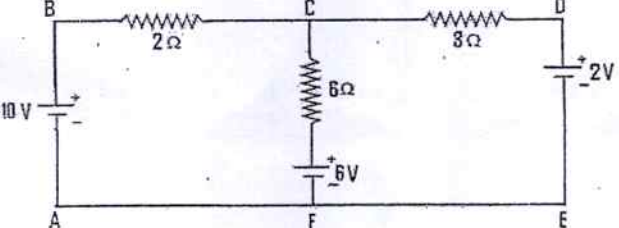
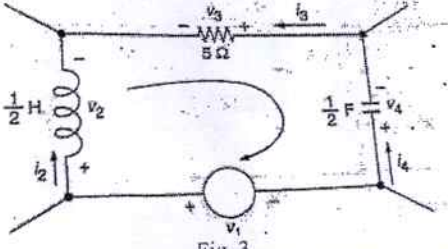
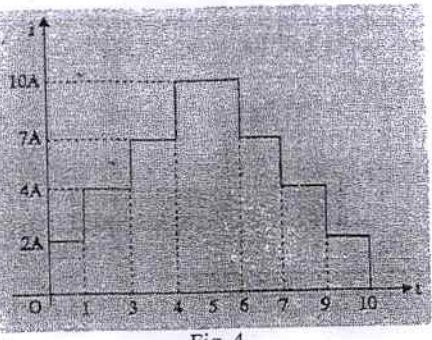
Semester: I  
Course Code: TEE 101

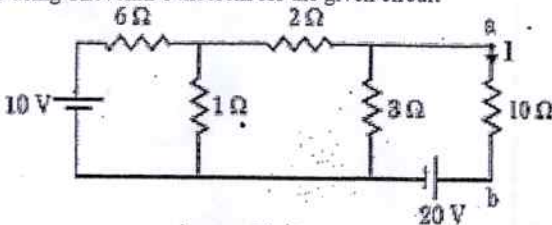
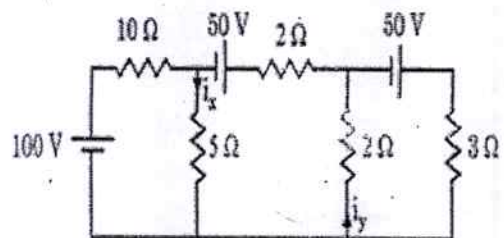
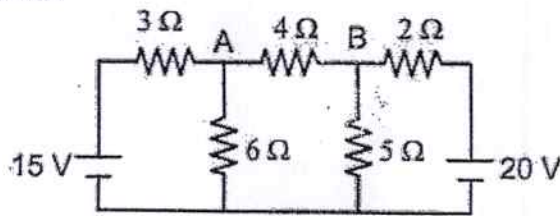
Time: 3 Hrs

Maximum Marks: 100

Note:-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

|     |                                                                                                                                                                                                                                                                                                                                         |     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q1  | (20marks)                                                                                                                                                                                                                                                                                                                               |     |
| (a) | Define a) Linear, and non Linear elements b) Active and Passive elements c) Current division and voltage division rule d) Mesh and loop                                                                                                                                                                                                 |     |
| (b) | <p>Determine the equivalent resistance between terminals A and B as shown in Fig. 1</p>  <p style="text-align: center;">Fig. 1</p>                                                                                                                    | CO1 |
| (c) | <p>Define the Kirchoff's Voltage law in electrical circuits. Find the voltage drop across 6 ohm resistance using mesh analysis as shown in Fig. 2</p>  <p style="text-align: center;">Fig. 2</p>                                                    |     |
| Q2  | (20marks)                                                                                                                                                                                                                                                                                                                               |     |
| (a) | <p>Determine the current <math>i_4</math> as shown in Fig. 3 when the following parameters are given:<br/><math>v_1 = 10 \sin t</math>, <math>i_2 = 4 \sin t</math>, and <math>i_3 = 2 \cos t</math></p>  <p style="text-align: center;">Fig. 3</p> | CO2 |
| (b) | <p>Prove that the current in purely Capacitive circuit leads its voltage by <math>90^\circ</math> and average power consumption in pure capacitor is zero.</p>                                                                                                                                                                          |     |
| (c) | <p>Find the average value, rms value, form factor and peak factor of the waveform shown in Fig. 4.</p>  <p style="text-align: center;">Fig. 4</p>                                                                                                   |     |

|     |                                                                                                                                                                                                                                                                                                     |     |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| Q3  | (20marks)                                                                                                                                                                                                                                                                                           | CO3 |
| (a) | <p>Give the statement of Thevenin's Theorem. Find the current in 10 ohm resistance using Thevenin's theorem for the given circuit</p>  <p style="text-align: center;">Fig. 5</p>                                  |     |
| (b) | <p>If <math>i_1 = 10 \sin \omega t</math>, <math>i_2 = 20 \sin (\omega t + 30^\circ)</math> and <math>i_3 = 15 \cos (\omega t - 30^\circ)</math>. Find the resultant current using phasor diagram.</p>                                                                                              |     |
| (c) | <p>Define the following terms for AC (alternating current) signal: (i) Peak Factor (ii) Form Factor (iii) Average Value (iv) RMS Value (v) Time period.<br/>An alternation voltage is given by <math>V = 230 \sin 314t</math>. Calculate frequency, maximum value, average value and RMS value.</p> |     |
| Q4  | (20marks)                                                                                                                                                                                                                                                                                           | CO4 |
| (a) | <p>Define the following terms of switchgear:<br/>(i) SFU<br/>(ii) MCCB<br/>(iii) MCB</p>                                                                                                                                                                                                            |     |
| (b) | <p>Explain different types of electrical wiring.</p>                                                                                                                                                                                                                                                |     |
| (c) | <p>What is the importance of switches in an electrical circuit? Give classification of switches.</p>                                                                                                                                                                                                |     |
| Q5  | (20 Marks)                                                                                                                                                                                                                                                                                          | CO5 |
| (a) | <p>Determine currents shown in Fig. 6 by nodal analysis</p>  <p style="text-align: center;">Fig. 6</p>                                                                                                           |     |
| (b) | <p>What is the importance of earthing in an electrical circuit? Define different methods of earthing.</p>                                                                                                                                                                                           |     |
| (c) | <p>Determine the currents supplied by each battery using star-delta transformation.</p>  <p style="text-align: center;">Fig. 7</p>                                                                              |     |